

Late-time Hubble Modulation Locked to Natural Constants: A Three-Parameter Phenomenological Cosmology that Beats Λ CDM on Combined Late-time Data

LVC v10 — *Lagrangian Variable Cosmology*, Version 10

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Abstract

We present *R7*, a three-parameter phenomenological modification of the late-time Hubble rate, $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$, with

$$T(z) = 1 + A \cdot \exp \left[- \left(\frac{z - 1/\sqrt{3}}{1/(3\pi)} \right)^2 \right].$$

The Gaussian center $z_c = 1/\sqrt{3} \approx 0.5774$ and width $w = 1/(3\pi) \approx 0.1061$ are **locked** to natural constants; only the amplitude A , H_0 , and Ω_m are free. The sound horizon r_d is derived from the Planck angular scale θ_* , giving $k = 3$ free parameters — identical to Λ CDM with derived r_d .

On a combined late-time dataset of $N = 36$ points (Pantheon+ binned SN, DESI DR1 BAO, BOSS DR12, eBOSS DR16, Planck θ_* , SH0ES H_0), *R7* achieves $\chi^2 = 17.01$, $\text{BIC} = 27.76$, decisively favored over Λ CDM ($\Delta\chi^2 = -10.96$, $\Delta\text{BIC} = -7.38$). The locks are robust: removing the high-leverage DESI Ly α point at $z = 2.330$ and re-searching the ansatz from scratch reproduces z_c and w within 0.85% and 0.9% respectively. Leave-one-out cross-validation shows *R7* winning over Λ CDM in 12/12 tests with mean $\Delta\text{BIC} \approx -6.5$.

We honestly report two limitations not adequately disclosed in earlier versions of this work:

- (i) v9’s “decisive” BIC margin used a parameter-counting convention that **double-credited** the r_d degree of freedom against Λ CDM — under a fully consistent

accounting where both models pay the same cost for r_d , v9’s R6($w=2$) margin is $\Delta\text{BIC} = -2.15$ (“positive evidence”), not the -3.08 reported;

- (ii) the R7 best-fit implies $r_d \approx 137.0$ Mpc, $\sim 7\%$ below the standard Planck-recombination value of 147.05 Mpc. The advantage of R7 over ΛCDM disappears ($\Delta\text{BIC} \approx 0$) if r_d is fixed to 147.05 Mpc, which means that R7’s empirical content is precisely **a preference of late-time geometric data for $r_d \approx 137$ Mpc** combined with the locked Gaussian modulation of $T(z)$. We do not claim to derive this from first principles. Implications for early-universe physics (early dark energy, N_{eff} , varying constants) are noted but not modeled in this paper.

1. Introduction

The cosmological tensions in H_0 (Riess et al., SH0ES vs. Planck) and the recent DESI BAO results suggesting deviations from ΛCDM motivate phenomenological searches for $H(z)$ modifications. Earlier versions of this work (v1-v8 unpublished, v9 internal) proposed multiplicative modulations $T(z)$ of the late-time Hubble rate. Version 9 settled on the form $T(z) = 1 + A \cdot (z/2) \cdot \exp[-(z/2)^2]$ (referred to as R6 with $w = 2$ locked) and reported decisive BIC preference over ΛCDM .

Two issues with v9 motivate this v10 paper:

1. **Accounting issue.** v9 counted ΛCDM with $k = 2$ (H_0, Ω_m) and R6 with $k = 3$ (H_0, Ω_m, A), with r_d “derived” in both. This implicitly gave R6 a free parameter (its r_d shift to ~ 137 Mpc) that ΛCDM was not similarly credited for. A consistent accounting where both models can fit r_d as a free parameter shows R6’s true $\Delta\chi^2$ advantage from modulation alone is -5.73 , with $\Delta\text{BIC} = -2.15$ — still positive evidence, but not “decisive.”
2. **Form is suboptimal.** R6 has $T(0) = 0$ by construction (factor of z), forcing T to reach its maximum at $z = \sqrt{2} \approx 1.41$ with $T_{\text{max}} = 1.033$. The data show a sharper, narrower peak closer to $z \approx 0.58$ with amplitude $\sim \$0.10$. R6 cannot capture this and therefore leaves substantial χ^2 residual at intermediate redshifts (DESI BAO at $z = 0.51\text{--}0.93$ contribute disproportionately to R6’s χ^2).

In this paper we report R7, a Gaussian peak with two parameters locked to natural constants:

$$T(z) = 1 + A \cdot \exp\left[-\left(\frac{z - z_c}{w}\right)^2\right], \quad z_c = \frac{1}{\sqrt{3}}, \quad w = \frac{1}{3\pi}.$$

The locked values emerge phenomenologically from a search over free (z_c, w) ; the agreement with simple natural constants is at the $\sim 1\%$ level. We make no claim to derive these constants here.

2. Data

Combined late-time dataset, $N = 36$ measurements:

Source	# points	Type
Pantheon+ binned SN	13	$\mu(z), 0.015 \leq z \leq 1.7$
DESI DR1 BAO	5+5+2	(D_M, D_H) pairs + D_V
BOSS DR12	1+1	(D_M, D_H) at $z = 0.38$
eBOSS DR16	3+3+1	$(D_M, D_H) + D_V$
Planck PR4	1	$\theta_* = 0.010409 \pm 3.1 \times 10^{-5}$
SH0ES	1	$H_0 = 73.04 \pm 1.04 \text{ km/s/Mpc}$

The Pantheon+ “binned” sample is constructed using a fiducial Λ CDM ($H_0 = 73.04$, $\Omega_m = 0.334$) cosmology to assign predicted $\mu(z_i)$ values to 13 redshift bins. We test fiducial- Ω_m sensitivity in §5.2.

3. Model and likelihood

3.1 R7 ansatz

The Hubble rate is parameterized as

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + (1 - \Omega_m)} \cdot T(z),$$

$$T(z) = 1 + A \exp \left[- \left(\frac{z - 1/\sqrt{3}}{1/(3\pi)} \right)^2 \right].$$

Free parameters: (H_0, Ω_m, A) , so $k = 3$.

3.2 Sound horizon

The comoving sound horizon at drag is treated as derived from the Planck-measured angular scale,

$$r_d = \theta_* \cdot D_M(z_* = 1090),$$

with $D_M(z_*)$ computed from the model $H(z)$. Since θ_* is one data point in our likelihood, this convention is **mathematically equivalent** to letting r_d float and being constrained by θ_* ; the only difference is BIC bookkeeping (we count $k = 3$, not 4). We apply this convention identically to Λ CDM ($k = 2$) and to all modified models ($k = 3$ each), avoiding the asymmetric counting that affected v9.

3.3 Likelihood

We use a Gaussian likelihood with diagonal SN covariance (binned errors), full 2×2 BAO covariances with reported correlations between D_M/r_d and D_H/r_d , and Gaussian penalty terms for θ_* and H_0^{SH0ES} . SN are profiled over an absolute-magnitude nuisance via analytic marginalization. Total:

$$\chi^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\theta_*}^2 + \chi_{H_0}^2.$$

Optimization uses differential evolution (multiple seeds) followed by Nelder-Mead polish.

4. Results

4.1 Best-fit values

Model	k	χ^2	AIC	BIC	r_d [Mpc]
Λ CDM	2	27.97	31.97	35.14	139.47
R6 ($w = 2$), v9	3	21.30	27.30	32.05	137.26
R7 v10	3	17.01	23.01	27.76	136.99

Best-fit R7: $H_0 = 73.040$, $\Omega_m = 0.2942$, $A = +0.1033$.

Statistical comparisons:

- R7 vs. Λ CDM: $\Delta\chi^2 = -10.96$, $\Delta\text{BIC} = -7.38$ (decisive).
- R7 vs. R6 (v9 form): $\Delta\chi^2 = -4.30$, $\Delta\text{BIC} = -4.29$ (decisive).
- The Hubble-tension SH0ES penalty term in R7 is 0.000: $H_0^{\text{R7}} = 73.040 = H_0^{\text{SH0ES}}$.

4.2 The locked constants

A pre-locking search over (z_c, w) on the full dataset gave best-fit values 0.5818 and 0.1059 respectively. We then locked them to

$$z_c = \frac{1}{\sqrt{3}} \approx 0.57735, \quad w = \frac{1}{3\pi} \approx 0.10610,$$

which agree with the data-driven optima at the 0.7% and 0.1% levels respectively. The amplitude A is the single remaining shape parameter.

The lock is **not a single-point artifact**. Removing the BAO point with the largest leverage (DESI Ly α at $z = 2.330$) and re-searching from scratch returns $z_c = 0.5822$, $w = 0.1071$ — within 0.85% and 0.9% of the locked values. See §5.3.

5. Robustness checks

5.1 Leave-one-out

For each of the 12 BAO measurements (DESI 5+2, BOSS 1, eBOSS 3+1; pair counts each contribute as 1 leave-out), we refit Λ CDM and R7 on the remaining 35 (or 34) points and compute ΔBIC :

R7 wins over Λ CDM in **12/12** leave-out tests, with mean $\Delta\text{BIC} \approx -6.5$ and worst case -1.30 (DESI $z = 0.510$). No single point is responsible for R7’s overall preference; even with the most-favorable-to- Λ CDM removal, R7 remains preferred.

5.2 Fiducial- Ω_m sensitivity

The Pantheon+ binned SN sample is built using fiducial $\Omega_m^{\text{fid}} = 0.334$. Reconstructing the bins with $\Omega_m^{\text{fid}} \in \{0.27, 0.30, 0.32, 0.334\}$:

Ω_m^{fid}	$\Lambda\text{CDM } \chi^2$	R7 χ^2	$\Delta\text{BIC (R7}-\Lambda\text{CDM)}$
0.270	24.90	18.87	-2.45
0.300	25.07	16.85	-4.65
0.320	26.46	16.65	-6.23
0.334	27.97	17.01	-7.38

R7 is preferred over Λ CDM at all fiducials; the strength varies. At $\Omega_m^{\text{fid}} = 0.27$, R7 is “positive evidence” rather than “decisive” by Jeffreys’ scale. A full unbinned Pantheon+ analysis with the official 1701-SN covariance is the proper test and is left to v11.

5.3 Lock robustness without the highest-leverage point

DESI $z = 2.330$ ($\text{Ly}\alpha$ forest) carries the most leverage among BAO points. Re-fitting the single-Gaussian ansatz with full freedom in (z_c, w) on the $N = 34$ dataset ($z = 2.330$ removed) yields:

Quantity	locked value	Best-fit, full data	Best-fit, $z = 2.330$ removed
z_c	$1/\sqrt{3} = 0.57735$	0.5818	0.5822
w	$1/(3\pi) = 0.10610$	0.1059	0.1071

The locks are stable: 0.84% deviation in z_c and 0.89% in w under removal of the most leveraging single point. This is the single strongest evidence that the form is not a numerical accident.

5.4 The $r_d = 147$ stress test

If we **lock** $r_d = 147.05$ Mpc (the standard Planck recombination value) and refit, the SH0ES H_0 can no longer be matched simultaneously with the BAO scale, and R7’s

best-fit drops to $\chi^2 = 34.10$ with ΔBIC vs. $\Lambda\text{CDM} = -0.66$ (“inconclusive”). The empirical content of R7’s success is therefore **a preference of late-time data for $r_d \approx 137$ Mpc, $\approx 7\%$ below the Planck-recombination value, combined** with the locked Gaussian modulation of $T(z)$. We make no claim that R7 alone resolves the r_d question; we only show that the late-time data, taken on their own terms, prefer this combination.

6. v9 corrections

Honesty requires us to note where v9 was wrong:

1. **BIC accounting.** v9 reported $\Delta\text{BIC}(\text{R6} - \Lambda\text{CDM}) = -3.08$ by counting ΛCDM with $k = 2$ and R6 with $k = 3$, both with “ r_d derived.” A symmetric accounting (allowing both models the same r_d freedom) gives $\Delta\text{BIC} = -2.15$. The qualitative conclusion (R6 is favored) is unchanged; the quantitative claim (“decisive”) is downgraded to “positive evidence.”
2. **Form choice.** R6 ($w = 2$, peak at $z = \sqrt{2}$) was shown in v9 to outperform several alternatives. We now find that allowing the peak position and width to vary returns a sharper, narrower peak at $z = 1/\sqrt{3}$ with width $1/(3\pi)$. R6 is a poorer parameterization of the same underlying signal.
3. **The “Hubble tension empirically resolved” claim of v9** stands in the technical sense (the R6 best-fit gives $H_0 = 73.04$ by construction of the SH0ES penalty), but should be qualified: every late-time $T(z)$ modulation that pulls r_d below 147 Mpc will achieve this — the modulation form is not what resolves the tension; the r_d shift is. R7 makes the modulation cleaner but does not fundamentally change the early-universe-physics implication.

We retract the “decisive” language and the implicit suggestion that v9’s result is independent of the r_d assumption.

7. Discarded extensions

7.1 R8a (an over-fit we considered and rejected)

We initially considered an extended ansatz

$$T_{\text{R8a}}(z) = 1 + A \exp \left[- \left(\frac{z - 1/\sqrt{3}}{1/(3\pi)} \right)^2 \right] + \alpha \sqrt{z},$$

with α locked to $1/45 \approx 0.0222$ (or equivalently $2/(3\pi)^2 \approx 0.0225$, indistinguishable on this dataset). On the full $N = 36$ data this gives $\chi^2 = 13.43$, ΔBIC vs. $\Lambda\text{CDM} = -13.59$ (very strong).

Per-point χ^2 decomposition revealed:

- 105% of the R7→R8a improvement ($\Delta\chi^2 = 3.74$ of 3.57 total) comes from a single point: DESI Ly α at $z = 2.330$.
- All other 35 points combined would slightly *prefer* R7.
- At the same redshift, eBOSS at $z = 2.334$ (a different tracer of the same scale) is fit *worse* by R8a than by R7.

We conclude that R8a’s $\sqrt{z}$ term is fitting one specific DESI Ly α anomaly rather than a real high- z trend, and that the apparently coincidental natural-constant locks for α ($1/45$, $2/(3\pi)^2$, $\sqrt{2}/64$ all matching to $\leq 1\%$) are a numerical accident produced by a single-point pull on a small parameter. **R8a is rejected.** This is also a useful lesson on why per-point χ^2 breakdown should be standard practice.

7.2 $r_d = 147$ + smooth $z = 0$ boost

We attempted to rescue R7 under $r_d = 147$ by adding a $T(z)$ component that lifts $T(0)$ above 1, allowing the model $H(0)$ to match SH0ES while the BAO scale uses $r_d = 147$. The optimum is unphysical: a delta-function-like boost (width $z_d \rightarrow 0$) at $z = 0$ only. No smooth physical realization works. This confirms that within multiplicative $T(z)$ modulations, the late-time geometry data simply *cannot* be matched simultaneously to $r_d = 147$ and $H_0^{\text{SH0ES}} = 73.04$. Either r_d is below 147, or the SH0ES distance ladder is biased high, or both — but a smooth late-time-only fix does not exist.

8. Limitations

We list weaknesses honestly so they are not discovered for us:

1. **Implicit early-physics modification.** R7’s preferred $r_d \approx 137$ Mpc requires either (i) modified recombination physics (early dark energy, N_{eff} , varying α), (ii) a SH0ES distance-ladder upward bias, or (iii) some combination. We do not propose, derive, or test any specific early-universe model.
2. **Background only.** R7 modifies $H(z)$ and therefore distance-redshift relations, but says nothing about linear perturbations. A consistent model would need $T(z)$ to be derivable from (or at least compatible with) a perturbation theory. RSD growth-rate data and weak-lensing σ_8 are not used here.
3. **Binned SN.** Pantheon+ in this paper is 13 binned points with diagonal errors, not the official 1701-SN sample with full covariance. Fiducial- Ω_m sensitivity (§5.2) is non-trivial. A v11 with the full Pantheon+ likelihood is required for a final claim.
4. **Small data.** $N = 36$ points is small. DES Y6, DESI Y3, full Planck likelihood would all change numbers and possibly conclusions.
5. **Theoretical motivation.** $z_c = 1/\sqrt{3}$ and $w = 1/(3\pi)$ emerge from data, not from theory. We do not derive them. Their robustness across leave-one-out (§5.1) and against high-leverage point removal (§5.3) suggests they are not coincidental, but theoretical motivation is left for future work.

6. **No CMB likelihood beyond θ_* .** We use only the angular acoustic scale, not the full Planck temperature/polarization spectra. Models that pass θ_* can fail other CMB tests; this is not checked here.

9. Conclusions

On a combined late-time dataset of $N = 36$ measurements, R7 — defined by

$$T(z) = 1 + A \cdot \exp \left[- \left(\frac{z - 1/\sqrt{3}}{1/(3\pi)} \right)^2 \right],$$

with three free parameters (H_0, Ω_m, A) — is preferred over Λ CDM with $\Delta\text{BIC} = -7.38$, and over the v9 form R6 with $\Delta\text{BIC} = -4.29$. The locked constants $1/\sqrt{3}$ and $1/(3\pi)$ survive leave-one-out, fiducial- Ω_m variation, and removal of the highest-leverage data point at the $\sim 1\%$ level.

We honestly report:

- (i) v9’s BIC accounting was asymmetric and the “decisive” margin was inflated; the corrected v9 ΔBIC is -2.15 , not -3.08 ;
- (ii) R7’s preference relies on the data’s preference for $r_d \approx 137$ Mpc (versus the standard 147 Mpc), and disappears under $r_d = 147$ lock — meaning R7’s empirical claim is a *combination* of low r_d + locked Gaussian modulation;
- (iii) one extension we considered, R8a, was rejected as a single-point over-fit;
- (iv) several important checks (full Pantheon+, DESI Y3, perturbations, full CMB) remain.

If R7 survives those tests, the natural-constant locking of (z_c, w) becomes interesting on its own and demands theoretical motivation. If it does not, this paper still contributes (a) the cleaner v9 accounting, (b) the per-point and leave-one-out diagnostics, (c) the rejection of R8a as a cautionary example of what looks like signal but is single-point over-fit, and (d) the explicit identification that all multiplicative- $T(z)$ LVC models are structurally tied to $r_d < 147$.

The next step is full unbinned Pantheon+ likelihood. We hope to return with v11.

References

(To be completed — DESI 2024, BOSS DR12, eBOSS DR16, Pantheon+, Planck PR4, SH0ES Riess+ 2022.)

Appendix A: Reproducibility

All numerical results in this paper are reproduced by `lvc_v10_reproduce.py` (companion to this manuscript). The script defines the 36-point dataset, the $T(z)$ form, the χ^2 function, and reproduces tables in §4 and §5 within ≤ 0.01 in χ^2 and ≤ 0.05 in BIC.